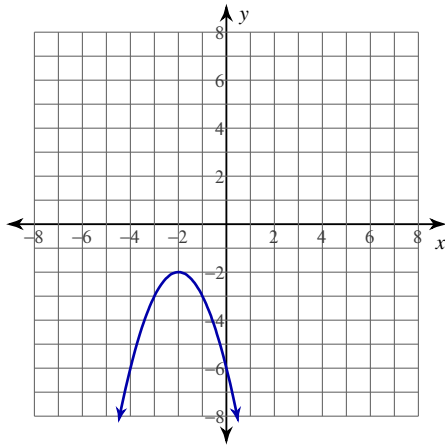


Horizontal Tangents

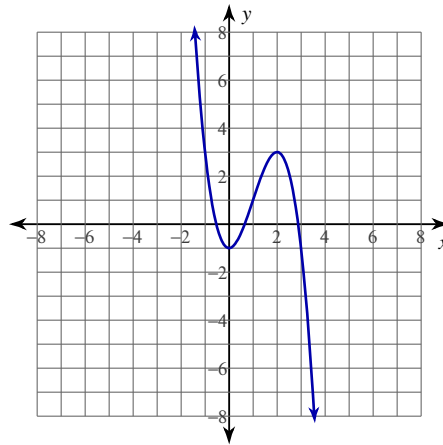
Date _____ Period _____

For each problem, find the points where the tangent line to the function is horizontal.

1) $y = -x^2 - 4x - 6$



2) $y = -x^3 + 3x^2 - 1$



3) $y = -x^3 + x^2 - 2$

4) $y = \frac{1}{x^2 - 1}$

For each problem, find the points where the tangent line to the function is horizontal. Indicate if no horizontal tangent line exists.

5) $y = x^3 - 2x^2 + 2$

6) $y = -x^3 + \frac{9x^2}{2} - 12x - 3$

7) $y = -\frac{2}{x-3}$

8) $y = -\frac{1}{x^2 + 1}$

9) $y = (-2x + 4)^{\frac{1}{2}}$

10) $y = -\csc(x); [-\pi, \pi]$